

AKSHAY ASHOK

M.Sc. Data Science & Artificial Intelligence | Saarland University

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PROFILE

Master's student in Data Science & AI (4th semester) who ships measured, reproducible ML: a first-author Springer publication in computer vision, an Arabic-to-German OCR/translation benchmark study for a Gov-Tech client, and a deployed NLP scoring service (FastAPI, Docker, CI). Experienced across the full pipeline, from dataset construction and model benchmarking to evaluation design, with LLM APIs and agentic tooling in daily use. Seeking Werkstudent, HiWi or internship roles in ML, data science or applied AI from August 2026.

EXPERIENCE

Student Consultant (AI / Data Science) · Lingomatch GmbH · Saarbrücken Apr 2026 – Jul 2026

- Built and benchmarked the ML backend of a document-triage feasibility study for a Gov/Justice-Tech language-services platform (UdS Innovation Center, Triathlon case-study programme): turning photos of Arabic documents into usable German gist translations within an 8 GB laptop-GPU budget.
- Evaluated Surya OCR on KITAB-Bench (per-sample CER/WER, confidence calibration); tested PaddleOCR as an independent second engine for disagreement-based flagging and documented why it fails on Arabic.
- Benchmarked Opus-MT vs. NLLB vs. TranslateGemma for translation; caught the licence issue (CC-BY-NC) ruling the best baseline out of production and recommended an Apache-2.0 model with comparable BLEU at 1/9 the memory.
- Designed confidence-based failure routing that flagged 4/4 catastrophic OCR failures, defining the pipeline's human-escalation path; presented the live demo in the final case-study presentation to the company's CEO.

github.com/Axwolf13/arabic-doc-triage-research · write-up: axwolf13.github.io/writing/arabic-ocr-benchmarks

Research Intern (Computer Vision) · National Institute of Technology, Puducherry Jan – Apr 2024

- Built a real-time yoga pose estimation pipeline with a hybrid MoveNet Thunder (training) + MoveNet Lightning (inference) architecture: 11 asanas at 98.8% test accuracy (98.4% verified on the shipped quantized model) at 30+ FPS on consumer hardware; curated the 3,435-image dataset end to end.
- First-authored a peer-reviewed publication in *Sādhanā* (Indian Academy of Sciences, Springer, 2025); released the code, the trained model and a live in-browser demo (TensorFlow.js) with a one-command verification script.

doi.org/10.1007/s12046-025-02788-w · live demo: axwolf13.github.io/demo

Web Development Intern · Zoho Corporation, Chennai, India Aug – Sep 2023

- Built a full-stack library management web app with a content-based book recommender (HTML/CSS, JavaScript, SQL).

SELECTED PROJECTS

DODI: deception scoring for Terms-of-Service documents, shipped as a live web service

- Deterministic index quantifying how hard a platform's ToS hides that "Buy now" means "revocable licence" (readability, licence-term ratio, red-flag clauses); validated against ToS/DR expert grades (Spearman $\rho = 0.54$, $n = 12$) and applied as a temporal study of 10 platforms across 2015–2024.
- Deployed end to end as a public scoring service: Dockerized FastAPI app with a test suite and CI on every push (GitHub Actions), live at dodi-web.onrender.com.

Shadow Correction: piracy as a market-correction signal · Data and Society, UdS · team of 3

- Quantified streaming-market fragmentation with the Herfindahl–Hirschman Index on Parrot Analytics demand-share data; built piracy-intent indices from Google Trends aligned with World Bank PPP-adjusted wages across 90+ countries, where the statistically-zero economic correlation isolates fragmentation, not affordability, as the driver.

MolCA reproduction: can an LLM read molecular graphs? · graduate seminar, in progress

- Critical verification of MolCA (EMNLP 2023) with Prof. Klakow's group: reproduce a downstream result, ablate the graph modality and test generalization beyond the authors' data (PyTorch, Hugging Face, Q-Former, LoRA).

EDUCATION

M.Sc. Data Science & Artificial Intelligence · Saarland University Oct 2024 – Present

Coursework: Statistics with R, Human-Computer Interaction, Data and Society. Current seminar: Machine Learning for Language Processing.

B.Tech. Artificial Intelligence & Data Science · Bannari Amman Institute of Technology Oct 2020 – Apr 2024

CGPA: 8.65/10. Thesis: LipNet-based visual speech recognition as a personal communication aid for hearing-impaired users.

TECHNICAL SKILLS & LANGUAGES

Languages & tools: Python, R, SQL, JavaScript; PyTorch, TensorFlow/Keras, Hugging Face Transformers, scikit-learn, spaCy, OpenCV, pandas, NumPy; FastAPI, Docker, CI/CD (GitHub Actions), Git, LaTeX.

LLM work: OpenAI & Anthropic APIs, prompt design, embedding-based retrieval, LLM output evaluation, agentic workflows (Claude Code, Cursor).

Methods: model benchmarking and evaluation design (BLEU, CER/WER, confidence calibration), classical ML & deep learning, mixed-effects models, hypothesis testing, time-series and event-study design, NLP and OCR pipelines, web scraping.

Spoken languages: English (C1), German (A2, targeting B1), Tamil (native), Hindi (basic).